Parabola: Formula Sheet

Conic and recognition

Definition: $PS = e \cdot PM$
Delta: $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$
Parabola: $\Delta \neq 0,\ h^2 = ab$
Ellipse / hyperbola: $h^2 < ab$ / $h^2 > ab$
Lines: $\Delta = 0$

Standard parabolas

$y^2 = 4ax$: $S(a, 0)$, $x = -a$
$y^2 = -4ax$: $S(-a, 0)$, $x = a$
$x^2 = 4ay$: $S(0, a)$, $y = -a$
$x^2 = -4ay$: $S(0, -a)$, $y = a$
LR: $4a = 2\times$ focus-directrix distance
Focal distance: $x_1 + a = a(1 + t^2)$
Shifted: $(y - k)^2 = 4a(x - h)$, focus $(h + a, k)$
Alternate:

$$\left(\frac{L_1}{\sqrt{l^2 + m^2}}\right)^2 = 4a\frac{L_2}{\sqrt{l^2 + m^2}}$$

Parametric form and chords

Point: $(at^2, 2at)$
Radius vector slope: $\frac{2}{t}$
Chord slope: $\frac{2}{t_1 + t_2}$
Chord: $2x - (t_1 + t_2)y + 2at_1t_2 = 0$
Through $(c, 0)$: $t_1t_2 = -\frac{c}{a}$
Focal chord: $t_1t_2 = -1$
Right angle at vertex: $t_1t_2 = -4$, through $(4a, 0)$
Focal chord length:

$$a\left(t + \frac{1}{t}\right)^2 = 4a\,\mathrm{cosec}^2\alpha$$

Chord on $y = mx + c$:

$$\frac{4}{m^2}\sqrt{a(1 + m^2)(a - mc)}$$

Position: $S_1 = y_1^2 - 4ax_1$ (< 0 inside)

Tangent

At point: $yy_1 = 2a(x + x_1)$
Parametric: $ty = x + at^2$
Slope form: $y = mx + \frac{a}{m}$
Contact: $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$, $m = \frac{1}{t}$
Meet of tangents: $(at_1t_2,\ a(t_1 + t_2))$
From (h, k): $hm^2 - km + a = 0$
Director circle: $x + a = 0$
Other curves: $c = -\frac{a}{m}$, $-am^2$, am^2

Normal

At point: $y - y_1 = -\frac{y_1}{2a}(x - x_1)$
Parametric: $y = -tx + 2at + at^3$
Slope form: $y = mx - 2am - am^3$
Foot: $(am^2, -2am)$, $m = -t$
Meets again: $t_2 = -t_1 - \frac{2}{t_1}$
Normals meet on curve: $t_1t_2 = 2$
Meet of normals:

$$\left(2a + a(t_1^2 + t_1t_2 + t_2^2),\ -at_1t_2(t_1 + t_2)\right)$$

From (h, k): $am^3 + (2a - h)m + k = 0$
Co-normal: $\sum m = 0$, $m_1m_2m_3 = -\frac{k}{a}$
Circle of feet: $2(x^2 + y^2) - 2(h + 2a)x - ky = 0$
Other curves: $c = 2am + am^3$, $2a + \frac{a}{m^2}$, $-2a - \frac{a}{m^2}$

Chord of contact, pole, diameter

Chord of contact / polar: $T = 0$
Mid-point chord: $T = S_1$
Pair of tangents: $SS_1 = T^2$
Chord of contact length:

$$\frac{\sqrt{(y_1^2 - 4ax_1)(y_1^2 + 4a^2)}}{a}$$

Triangle area:

$$\frac{(y_1^2 - 4ax_1)^{3/2}}{2a}$$

Pole of $lx + my + n = 0$:

$$\left(\frac{n}{l}, -\frac{2am}{l}\right)$$

Diameter: $y = \frac{2a}{m}$

Highlights

H.M. of focal segments: $\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a}$
Focal chord tangents: $\perp$, meet on directrix
Angle at focus (tangent to directrix): $90°$
Foot of $\perp$ from S: on $x = 0$
Image of S in tangent: on directrix
T, G on axis: $ST = SG = SP$
Triangle area: $a^2|(t_1 - t_2)(t_2 - t_3)(t_3 - t_1)|$ $(2:1)$